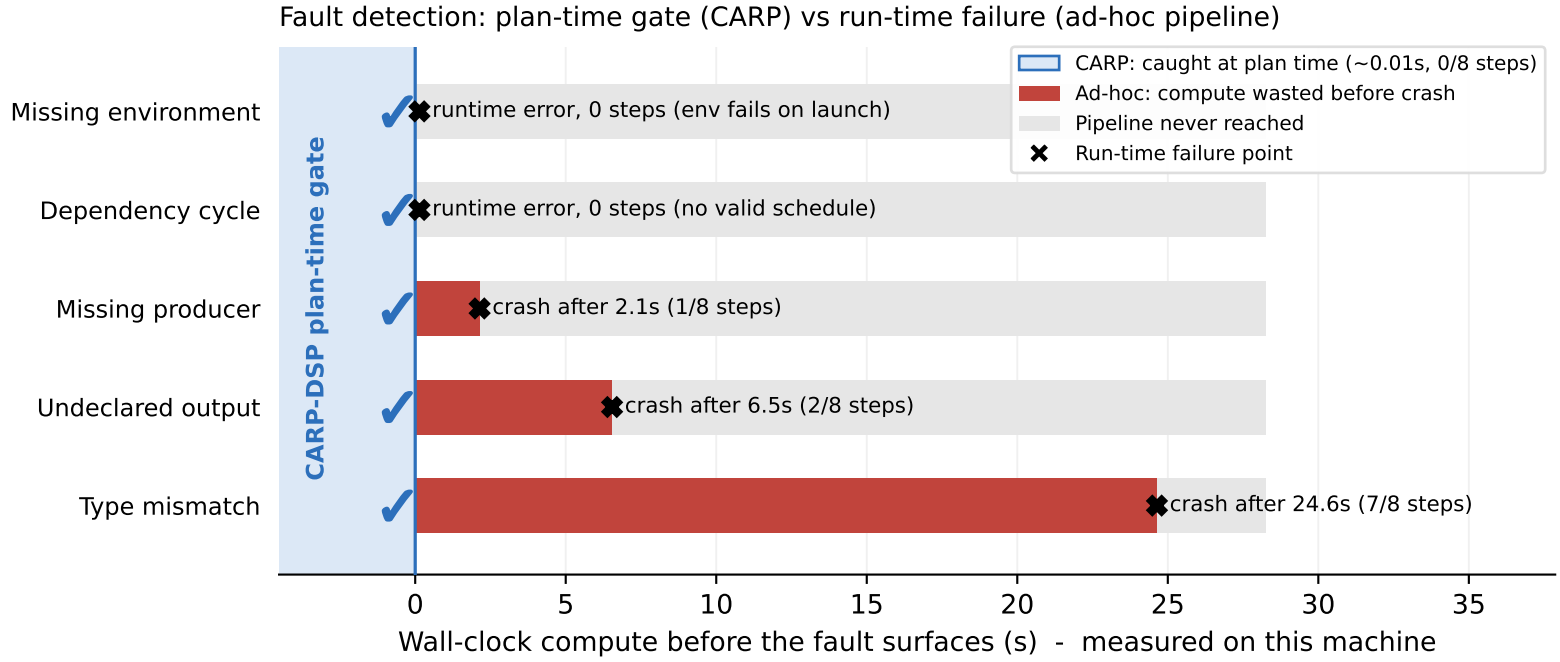
Fault detection: plan-time gate (CARP) vs run-time failure (ad-hoc pipeline)
CARP: caught at plan time (~0.01s, 0/8 steps)
Ad-hoc: compute wasted before crash
Pipeline never reached
Run-time failure point
CARP-DSP plan-time gate
Missing environment
runtime error, 0 steps (env fails on launch)
Dependency cycle
runtime error, 0 steps (no valid schedule)
Missing producer
crash after 2.1s (1/8 steps)
Undeclared output
crash after 6.5s (2/8 steps)
Type mismatch
crash after 24.6s (7/8 steps)
0
5
10
15
20
25
30
35
Wall-clock compute before the fault surfaces (s) - measured on this machine